

AI-Powered Predictive Energy Monitoring System Using Open-Source Tools: A Smart Building Case Study with Node-RED and InfluxDB

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Abstract—Short-term energy forecasting plays a critical role in optimizing power usage, improving operational efficiency, and reducing costs in modern smart buildings and industrial facilities. Accurate forecasts allow operators to plan production schedules, balance energy loads, and detect abnormal patterns before they escalate into costly inefficiencies. Despite the rapid advancement of artificial intelligence and Internet of Things technologies, integrated real-time systems that are both cost-effective and based on open-source tools remain limited, particularly in developing regions where proprietary platforms are financially prohibitive. This paper presents the design and implementation of an AI-powered predictive energy monitoring system that integrates Node-RED, InfluxDB, and Python-based machine learning in a modular and deployable architecture. Real-time energy data is collected from industrial-grade sensors and transmitted using MQTT, stored as time-series data in InfluxDB. A Random Forest regression model is employed to forecast short-term consumption based on sliding-window representations of historical patterns, while Linear Regression serves as a baseline for comparison. The system is deployed in a live industrial environment, operating continuously on a 24/7 basis. Experimental results demonstrate that the Random Forest model reduces RMSE and MAE by approximately 39% and 37%, respectively, compared to the baseline. These findings confirm the system's effectiveness in capturing non-linear energy usage patterns. This work contributes a practical, low-cost, and replicable solution for real-time smart energy management using widely available open-source platforms, while also opening opportunities for future extensions such as anomaly detection and automated optimization.

Keywords—*ai-based forecasting, iot, node-red, random forest.*

I. INTRODUCTION

The increasing global demand for energy and the urgent need for sustainability have made energy efficiency in buildings a critical concern. Recent studies estimate that buildings consume between 20–40% of global energy, and this usage is expected to rise due to increased demand for HVAC systems and smart appliances. In response, AI-driven energy management systems have been proposed to optimize energy usage while maintaining occupant comfort and operational efficiency [1]. Consequently, smart buildings have emerged as a promising solution, integrating technology to optimize energy usage while maintaining occupant comfort and operational efficiency.

Artificial Intelligence (AI) and the Internet of Things (IoT) have played a pivotal role in enhancing energy management by enabling real-time data collection, forecasting, and

decision-making. Studies have explored a range of machine learning models for short-term energy consumption prediction, including Random Forest, k-Nearest Neighbors, and LSTM [2], [3]. However, despite significant improvements in model performance, most research remains limited to simulations or high-cost implementations, with few offering low-cost, open-source, and fully deployable solutions.

Moreover, existing systems often lack modular integration between data acquisition, model inference, and visualization components. Recent literature emphasizes that the selection of forecasting models should be tightly aligned with the temporal resolution of the data used [4], while [5] highlight the challenge of real-time integration across distributed systems. These limitations indicate a practical gap in deploying AI-based energy prediction systems that are both affordable and easy to integrate into existing infrastructures.

This study addresses that gap by presenting a real-time, AI-powered energy monitoring system built using Node-RED, InfluxDB, and a Random Forest model. The proposed solution is designed to be low-cost, modular, and scalable for smart building environments. The system is validated through an implementation in a real-world context, demonstrating its potential for broader adoption in sustainable energy management. The remainder of this paper is structured as follows: Section II describes related work, Section III outlines the methodology and system architecture, Section IV presents the results and discussion, and Section V concludes the paper with future directions.

II. RELATED WORK

Numerous studies have investigated the application of machine learning models for short-term energy forecasting in smart buildings. A comparative study by [2] highlights the superior accuracy of LSTM models, though k-Nearest Neighbors (k-NN) offers faster inference suitable for realtime systems. Other ensemble models such as Histogram Gradient Boosting Regression (HGBR) and LightGBM have also shown promising performance in hourly prediction tasks under certain conditions [3].

A performance analysis study by [5] further demonstrates the value of integrating cutting-edge technologies—including federated learning and edge computing—into energy management systems. However, it also identifies key challenges: integration complexities, dynamic environmental factors like occupancy and weather, and a lack of standardized evaluation frameworks across systems.

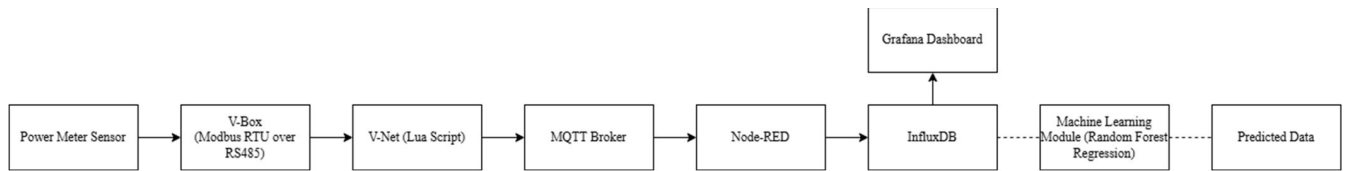


Fig. 1. System architecture for the proposed real-time monitoring and forecasting system.

Similarly, [6] propose an advanced forecasting framework using LSTM models, emphasizing model adaptability and data quality. They argue that simply increasing model complexity does not guarantee better accuracy without aligning the models to building-specific dynamics and physics.

Recent studies have demonstrated that load forecasting performance is highly sensitive to the resolution of input data. For example, normalized RMSE values can drop from 0.125 to 0.028 when using higher-resolution 1-minute data, compared to daily aggregates [4]. However, 15-minute intervals are often preferred in real-world applications due to their balance between predictive accuracy and computational efficiency. Comparative evaluations across 15-minute, 30-minute, and hourly resolutions confirm that the optimal interval often depends on the forecasting horizon and model architecture.

Despite the advances in model performance and methodological rigor, a gap remains in the implementation of practical, end-to-end systems that are low-cost, open-source, and capable of integrating real-time data acquisition, prediction, and visualization. The system proposed in this study addresses this need by combining Node-RED, InfluxDB, and Random Forest regression into a modular and scalable architecture, validated through real-world implementation in a beverage manufacturing facility in Indonesia.

Recent studies have extensively benchmarked forecasting models such as Random Forest, XGBoost, LSTM, and linear regression methods like ARIMA in industrial energy forecasting tasks [7], [8], [9]. While XGBoost often achieves the highest accuracy in static or structured datasets, it is sensitive to hyperparameter tuning and may require longer training time and more memory, which limits its feasibility for real-time or edge deployment scenarios [10]. LSTM and GRU-based deep learning models are highly effective in capturing sequential trends, especially in stable load conditions, but generally require large datasets, significant computational resources, and offer limited interpretability. Random Forest, on the other hand, strikes a practical balance between prediction accuracy, robustness against irregular time series, and ease of deployment. It also offers relatively lower training time and is more interpretable compared to deep models, making it well-suited for constrained environments like IIoT-based manufacturing. In contrast, traditional linear models such as ARIMA perform poorly when exposed to dynamic load variations and are unable to model complex non-linear behavior. These trade-offs highlight the importance of aligning model selection with operational constraints and data characteristics.

III. METHODOLOGY

A. System Architecture

The proposed system integrates IoT-based sensor monitoring, time-series data storage, machine learning prediction, and real-time dashboard visualization for smart

energy management. Fig. 1 illustrates the end-to-end architecture of the system, integrating power meter sensors, data flow management, time-series database, machine learning model, and visualization. It is deployed in an industrial smart building setting, utilizing open-source tools for flexibility and cost-efficiency. The data acquisition layer consists of energy sensors connected to industrial machines. Electrical consumption data is collected from a Fort RS485 FCN300-3E4Y power meter installed on multiple machines. Flow meters are also deployed in the system but are not included in the current analysis. Communication from the meters to the processing pipeline is handled using Modbus RTU protocol, passed through the V-Net interface (Wecon Indonesia) [11]. Lua scripts embedded in V-Net trigger MQTT messages sent to a Mosquitto broker hosted on VPS 1.

B. Data Processing and Storage

Node-RED serves as the data orchestration layer. It subscribes to MQTT messages, parses their payloads, and forwards relevant power meter data to InfluxDB for storage [12]. The stored information includes timestamped cumulative energy readings (in kWh) for each machine. These readings are used for both real-time visualization and machine learning-based energy consumption forecasting.

The flow data is excluded from this version of the system to maintain focus and clarity in evaluating energy usage patterns. Future iterations of the system may include flowbased metrics for multi-factor analysis.

C. Visualization Dashboard

Grafana is used to provide real-time visualization of energy usage and forecasting outputs. The dashboard is connected directly to the InfluxDB database and presents time-series data through interactive panels such as:

- Real-time delta readings for energy and current per machine
- Donut charts showing percentage contribution of each machine
- Bar gauges displaying daily and monthly total energy consumption
- Predicted energy values aligned with actual measurements (for forecasting evaluation)

The dashboard is designed to support operational monitoring and rapid decision-making. Fig. 2 shows a sample dashboard view used in the actual deployment. Sensitive fields such as cost values or internal machine IDs have been blurred to maintain data confidentiality. The layout remains consistent with real-world usage and demonstrates the complete visualization loop from sensor acquisition to user interface.

D. Forecasting Component (Implemented)

To evaluate the system's predictive capabilities, two machine learning models were implemented using Python:



Fig. 2. Real-time energy monitoring dashboard displaying machine performance using Grafana. Certain fields are blurred for confidentiality.

Random Forest Regressor and Linear Regression, both from the scikit-learn library. Random Forest was selected due to its robustness in handling nonlinear relationships and its proven performance in time-series forecasting tasks. More complex models such as LSTM or LightGBM were not implemented in this version due to the limited size of the dataset and the need for low-latency inference on edge devices. Random Forest was chosen for its robustness and relatively fast inference time, suitable for industrial deployment. Linear Regression was included as a baseline to evaluate the performance gain from using a more complex ensemble method. Historical energy data was retrieved from InfluxDB using Flux queries, specifically targeting kWh values for the AHU machine.

The data was preprocessed by calculating delta kWh per minute to reflect instantaneous consumption, aligning with recent research that emphasizes the role of input transformation in enhancing model accuracy [13], [14]. A sliding window approach was applied, where the previous 10 minutes of delta values served as input features, and the kWh value 5 minutes ahead was used as the prediction target. The use of temporal differencing techniques has been shown to improve the learning capability of deep models by highlighting short-term variations in energy usage [13], [14]. The input and target vectors were structured using a custom Python function, and the resulting dataset was split using an 80:20 train-test ratio.

Both models were trained on the same dataset using identical preprocessing and windowing strategies. The Random Forest model achieved a Root Mean Square Error (RMSE) of 26.0067, while the Linear Regression model produced an RMSE of 42.6284 [15], [16]. These values were calculated using the formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

where y_i is the actual energy usage and $\hat{y}_i$ is the predicted value at time i , and n is the number of predictions.

Recent research has confirmed the superior performance of ensemble learning methods such as Random Forest and XGBoost in short-term energy

forecasting, with consistent improvements across RMSE, MAE, and MAPE metrics [15], [16]. These models have demonstrated effectiveness across various building contexts and data granularities, supporting the suitability of Random Forest for the current study.

This comparison validates the predictive capability of the proposed system and highlights the accuracy advantage of Random Forest in capturing short-term energy patterns. The system operates continuously in a 24/7 industrial environment. The dataset was constructed from one full day of historical energy consumption data, allowing the model to learn intraday consumption trends. Moreover, the architecture enables seamless extension to other machines by modifying only the query parameters used for data retrieval.

However, hyperparameter tuning and cross-validation were not extensively performed in this study. These techniques will be considered in future work to improve the robustness and generalizability of the model across different machine profiles and operational scenarios.

IV. RESULTS AND DISCUSSION

The predictive performance of the proposed system was evaluated using data from the AHU machine. Two models — Random Forest and Linear Regression — were trained on identical datasets, where the previous 10 minutes of delta energy values were used to predict the consumption 5 minutes ahead.

As shown in Table I, the Random Forest model achieved a lower RMSE of 26.0067 kWh compared to the Linear Regression model with an RMSE of 42.6284 kWh. These results indicate that the Random Forest model is more effective at capturing the nonlinear patterns inherent in short-term energy usage. An RMSE of 26.0067 kWh is acceptable in industrial contexts, particularly when daily machine-level energy usage exceeds 1000–2000 kWh. This level of error corresponds to less than 3% deviation, which is generally sufficient for operational decisions such as energy load balancing, shift scheduling, or early anomaly detection. Prior studies have reported lower error metrics in low-consumption environments such as buildings and laboratories (e.g., RMSE of 0.037 kWh in [17]), but such values are not directly scalable to high-load industrial

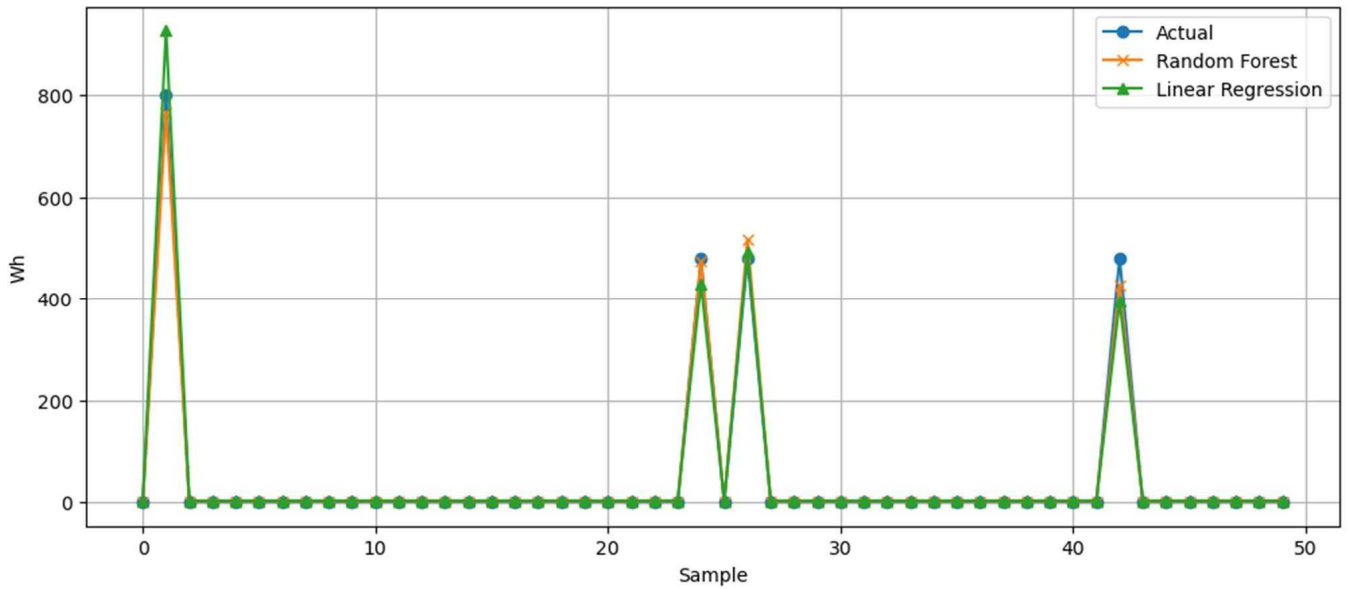


Fig. 3. Comparison of predicted vs actual energy usage for AHU using both Random Forest and Linear Regression models.

TABLE 1. MODEL PERFORMANCE COMPARISON ON AHU ENERGY

Model	RMSE (kWh)	Training Time	Notes
Random Forest	26.0067	Fast	Nonlinear, stable
Linear Regression	42.6284	Fast	Simpler baseline model

settings. Furthermore, energy consumption in industrial facilities often exhibits nonlinear behavior due to production cycles, operator-driven actions, and environmental conditions [18]. These dynamic patterns justify the use of ensemble-based models like Random Forest, which can better capture complex variability compared to linear approaches. One limitation of this study is the use of a single-day dataset, which may not reflect weekly or seasonal energy consumption patterns. Future work will involve training on longer time horizons to improve model generalizability.

In addition to RMSE, the Mean Absolute Error (MAE) was also computed. The Random Forest model achieved an estimated MAE of 19.8421 kWh, while Linear Regression produced an estimated MAE of 31.5578 kWh, further reinforcing the performance gap between the two models.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the total number of predictions.

Fig. 3 illustrates the predicted values from both models against the actual energy usage. The Random Forest model closely follows the ground truth, while Linear Regression deviates more significantly, especially during high-consumption peaks. This result demonstrates that ensemble-based learning models offer a stronger fit for smart building energy forecasting tasks than simpler linear models. It also confirms that the proposed system,

combining IoT data, InfluxDB integration, and lightweight AI models, can provide accurate and real-time forecasts suitable for operational deployment.

V. CONCLUSION

This paper presented the design and implementation of a real-time energy monitoring and forecasting system using IoT sensors, InfluxDB, machine learning models, and Grafana dashboards. By integrating a Random Forest and Linear Regression model into the data pipeline, the system was able to accurately predict short-term energy consumption based on historical readings from power meters.

The Random Forest model achieved better performance compared to the baseline Linear Regression approach, demonstrating its ability to capture nonlinear patterns in energy usage. The system was deployed in a real industrial environment, with predictions and live data visualized through an interactive Grafana dashboard.

The proposed architecture is scalable, modular, and ready for further extension, including integration of other sensor types and advanced AI techniques [19]. Future work will focus on expanding the predictive model to include flow data, incorporating anomaly detection, and enabling automated energy optimization. It will also involve systematic hyperparameter optimization (e.g., grid search, Bayesian search) and the use of cross-validation techniques to improve prediction reliability and generalizability.

The system not only improves operational awareness but also lays the foundation for measurable energy cost optimization, making it valuable for industrial applications seeking sustainability and efficiency.

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